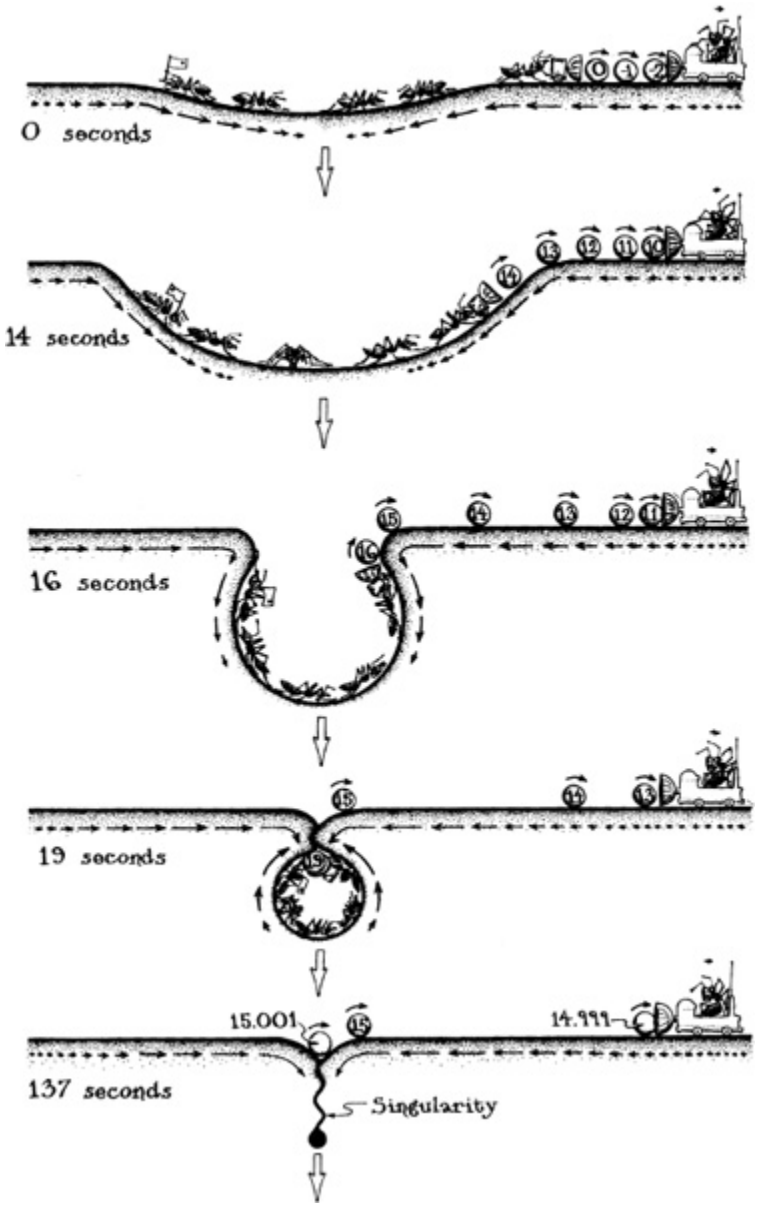


6.6 Collapsing rubber membrane populated by ants provides a fanciful analogue of the gravitational implosion of a star to form a black hole. [Adapted from Thorne (1967)]

The membrane's surface contracted faster and faster as the collapse proceeded. As a result, the signal balls, which were uniformly spaced in time when dispatched by the trapped ants, were received by the astronomer ant at more and more widely spaced time intervals. (This is analogous to the reddening of light from an imploding star.) Ball number 15 was dispatched 15 seconds after the collapse began, at the precise moment when the trapped ants were being sucked through the membrane's critical circumference. Ball 15 stayed forever at the critical circumference because